Rivan Jarjes

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Education

Toronto Metropolitan University Sep 2024 – May 2028
Bachelor of Science (Honours) in Computer Science – Co-op *Toronto, ON*

- **GPA:** 3.68/4.33
- **Awards:** NSERC Undergraduate Student Research Award (2025, 2026); Dean's List (2024–25, 2025–26)
- **Coursework:** Data Structures & Algorithms, Operating Systems, Software Engineering, Database Systems

Technical Skills

Languages: Python, Go, TypeScript, JavaScript, C#, SQL
Frameworks: React, Next.js, FastAPI, .NET, Ruby on Rails
Cloud & Infrastructure: GCP, AWS, Kubernetes, Docker, Cloud Run, Pub/Sub
Databases & Tools: PostgreSQL, Redis, Git, GitHub, CI/CD, Claude Code, Linux, Jira, pytest, xUnit

Experience

Software Developer Intern Sep 2025 – Dec 2026
Geotab *Oakville, ON*

- Built and deployed a Kubernetes-based Python backend for AI coding agents, enabling 82 developers to launch Jira-driven implementation, code-review, and testing workflows.
- Reduced CI build-stage runtime by 16.1% (3.2 min/run) across 4 .NET repositories by enabling NuGet package caching and validating gains across build jobs.
- Resolved recurring API encoding/decoding defects across a 61-issue backlog by centralizing request/response handling in Axios interceptors and adding regression tests while preserving backward compatibility.
- Expanded React production telemetry to capture 53 previously untracked errors in 2 months by serializing object payloads and filtering rejection noise.

Research Assistant (Software Engineering & Applied AI, Part-Time) May 2025 – Present
University of Toronto *Toronto, ON*

- Led full-stack development of a learning platform used by 400+ students across 5 university courses, building REST and WebSocket APIs with FastAPI, PostgreSQL, Next.js, and React.
- Architected distributed, event-driven services using GCP Cloud Run, Pub/Sub, Redis, and PostgreSQL, processing 1,696 asynchronous jobs with 99.9% completion.
- Implemented resumable, versioned workflows with Redis-backed state and distributed locking, supporting peak loads of 425 jobs/day.
- Engineered a real-time classroom presentation system used in 89 live sessions, synchronizing interactions across 21 displays for up to 50 concurrent participants.

Web Developer Apr 2025 – Apr 2026
Metropolitan Game Studios — TMU Student Club | ***metropolitangamestudios.ca*** *Toronto, ON*

- Directed development of Metropolitan Game Studios' public website using Next.js, React, and Tailwind CSS, creating a responsive platform for team profiles, events, and game showcases.

Projects

Event-Driven Product Taxonomy Classifier | *Ruby on Rails, Go, Redis* | GitHub Nov 2025

- Engineered an event-driven Rails pipeline using Sidekiq/Redis to asynchronously dispatch merchant image classification to a Go microservice hosting Qwen3-VL.
- Built constrained-generation tooling that converts Shopify's product taxonomy into a GBNF grammar, constraining the vision model to return schema-valid product categories and JSON.

Web-Based RISC Assembly Simulator | *TypeScript, Next.js, React* | GitHub | Live Demo Jan 2025

- Developed a browser-based ARC assembly simulator used by TMU Computer Architecture students, implementing instruction execution, registers/memory state, breakpoints, syntax validation, and interactive debugging.